

Unkey

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Key Metrics

Pages scanned: 114

Report type: Premium Executive Audit

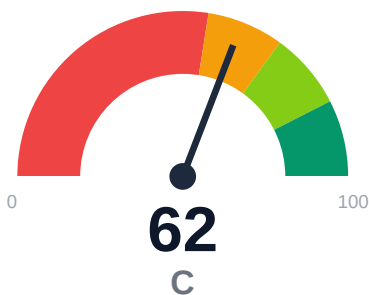
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 17
- API usage-doc coverage is low: 13.16% (33 API pages detected)
- Example reliability estimate is low: 0.0%

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit of <https://unkey.com/docs> covered 114 pages with 100.88% crawl coverage and 91.5% overall confidence. Critical issues include 17 confirmed broken internal documentation links requiring immediate remediation. API reference coverage is severely deficient at 13.16% despite 33 API pages detected, indicating incomplete documentation of available endpoints. Example code reliability scored 0.0%, representing a complete absence of validated or testable code samples. Freshness metadata exists for 76.32% of pages, while SEO/geo issues affect 9.65% of content.

External posture: SEO/GEO issue rate **9.7%**, public API coverage **13.2%**, broken links **17**.

62 Audit Score	C Grade	High Risk Band
114 Pages Scanned	17 Broken Links	9.7% SEO Issue Rate

Per-Site Breakdown

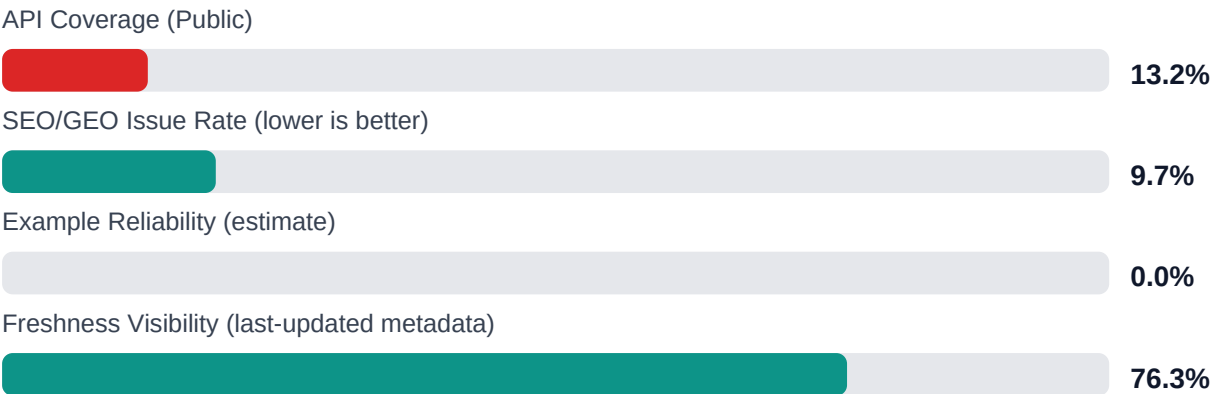
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://unkey.com/docs	114	17	13.2	0.0	9.7

Industry Benchmark Comparison

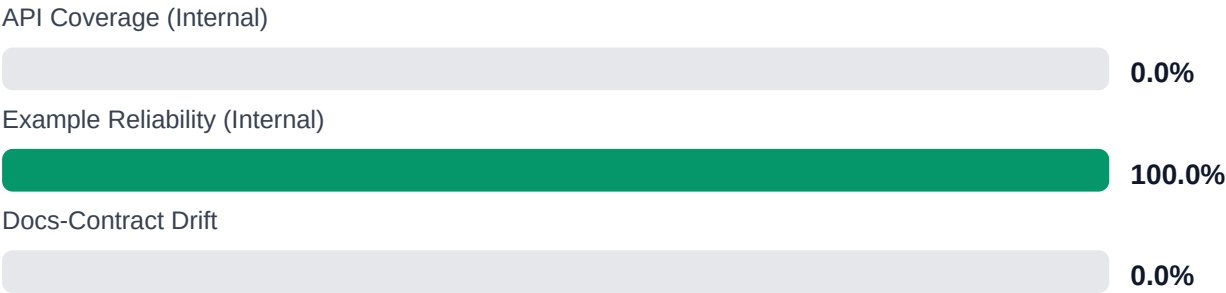
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-30
Industry average (developer docs)	85	-23
Minimum acceptable	70	-8
Your score	62	-

Gap to industry average: **23 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	\$249,513 annual exposure Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	\$431,728 annual exposure Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 17 broken internal documentation links through automated link validation in CI/CD pipeline [impact: critical, effort: low]

2. Expand API reference documentation from 13.16% to minimum 80% coverage by documenting missing endpoints, parameters, and response schemas [impact: critical, effort: high]
3. Implement executable code examples with automated testing to achieve minimum 60% example reliability score [impact: critical, effort: high]
4. Add last-updated metadata to remaining 23.68% of pages and establish quarterly content review process [impact: medium, effort: low]
5. Remediate SEO/geo issues on affected 9.65% of pages (meta descriptions, canonical tags, language declarations) [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage at 100.88% with all 114 discovered pages successfully examined
- + High overall audit confidence at 91.5% with strong link verification (95%) and crawl reliability (95%)
- + Single-product topology provides clear documentation structure and navigation model
- + No repository navigation link breakage (0 repo_broken_links_count), indicating healthy source control integration
- + Majority of pages (76.32%) include last-updated metadata supporting freshness tracking

Risks

- 17 confirmed broken internal documentation links create poor user experience and erode trust in documentation quality
- API reference coverage at 13.16% leaves 86.84% of API surface undocumented, blocking developer adoption and integration
- Example reliability at 0.0% indicates no working code samples, forcing users to debug documentation errors
- Zero validated examples increase support burden and slow time-to-first-success for new developers
- 23.68% of pages lack freshness metadata, obscuring content staleness and maintenance gaps
- SEO/geo issues on 9.65% of pages may reduce discoverability in search engines and international markets
- Low API confidence score (85%) suggests potential gaps in automated API endpoint detection or documentation structure

Limitations

- * External audit cannot verify technical accuracy of API documentation content, parameter types, or response examples against actual API behavior
- * Example code reliability scoring depends on detectable patterns; custom or embedded code samples may not be fully assessed
- * API coverage detection relies on documentation structure conventions; non-standard reference page formats may undercount actual coverage
- * Audit cannot assess documentation completeness for proprietary features, internal tools, or undisclosed API capabilities
- * Freshness metadata presence does not guarantee content accuracy or reflect actual product changes since last update

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.